

CSE-4878

Machine Learning and Data Mining Lab

Integrated 15-Week Laboratory Manual - 30 Classes

REVISED EDITION

Incremental progression: Python → Data Analysis → Preprocessing → Data Mining → Machine Learning → Text Analytics → Final Project Evaluation

Course Information	Details
Course Code	CSE-4878
Course Title	Machine Learning and Data Mining Lab
Credits	1
Laboratory Pattern	15 weeks × 2 classes per week = 30 classes
Prerequisite	CSE-3636 (Artificial Intelligence Lab)
Co-requisite	CSE-4877 (Machine Learning and Data Mining)
Primary Environment	Python 3.x, Jupyter Notebook / Google Colab
Final Project Assessment	SEE - 40 marks + 10 marks on report

Revision Note

This revision preserves the laboratory experiment families from the existing CSE-4878 lab manual and aligns the final five classes strictly with the SEE/project evaluation rubric in the project-format document. Classes 1–25 are instructional laboratories and project-readiness preparation. Classes 26–30 are reserved exclusively for final project evaluation; no new project-development content is assigned in those five classes.

1. Source Alignment and Design Principle

The existing laboratory material covers statistical analysis, preprocessing, association-rule mining, regression, classification, neural networks, clustering and ensemble learning. The project-format document requires a text-data research project, at least two algorithms, evaluation evidence, an IEEE-style paper, a GitHub repository and a five-minute presentation.

The course is therefore sequenced so that students do not encounter text analytics or machine-learning experiments before acquiring the necessary Python, numerical computing, data handling, statistics and preprocessing skills.

The earlier integrated version incorrectly distributed final-project development and evaluation across the last weeks. In this revision, the final five classes correspond one-to-one with the five SEE rubric components.

2. Mapping of CLO-PLO-DL-KP-EP-EA

#	CLOs	PLOs	DL	KP	EP	EA	Teaching Learning Strategy (TLS)	Assessment Strategy (AS)
CLO1	Demonstrating statistical data analysis with modern tools.	PLO5	C2	6	1,3,5		Lecture, Class discussion, Assignment, Lab work, Note	Exam, Lab performance
CLO2	Apply data preprocessing techniques to improve model performance.	PLO2	C2	2,3	1,3,4		Lecture, Class discussion, Assignment, Lab work, Note	Exam, Lab performance
CLO3	Design and evaluate machine learning models using appropriate algorithms and metrics.	PLO4	C3	8	1,2,3		Lecture, Class discussion, Assignment, Lab work, Note	Exam, Lab performance
CLO4	Analyze fairness, bias, and legal aspects in algorithmic decision-making.	PLO6	C4	7	1,3,7		Lecture, Class discussion, Assignment, Lab work, Note	Exam, Lab performance
CLO5	Demonstrate the ability to assess the societal and environmental impacts of the systems and to document project design, implementation, and outcomes through well-structured reports and presentations.	PLO7	C4	7	1,3,7		Lecture, Class discussion, Assignment, Lab work, Note	Project report

Note: DL: Domain/level in taxonomy, KP: Knowledge Profile, EP: Attribute of Complex Engineering Problem, EA: Attribute of Engineering Activities, Learning Domains (C: Cognitive, A: Affective, P: Psychomotor).

3. Laboratory Operating Rules

- Use Jupyter Notebook or Google Colab for laboratory work.

- Every notebook should contain: Objective, Learning Outcomes, Background/Theory, Required Libraries, Dataset, Implementation, Results/Visualization, Interpretation, Limitation/Error Analysis, Conclusion and Viva Preparation.
- From the preprocessing laboratories onward, maintain a correct separation between training and testing data and document every preprocessing decision.
- When a from-scratch implementation is requested, first implement the core idea on a small dataset and then verify the result using a standard library where appropriate.
- Students should retain experiment logs, figures and tables because these provide evidence for later project work.
- Large datasets should be referenced through a stable source rather than unnecessarily committed to GitHub.
- Final project evaluation in Classes 26–30 is assessment-only. Students are not expected to learn new project-development techniques during those classes.

4. Recommended Software Environment

- Python 3.x
- Jupyter Notebook or Google Colab
- NumPy, Pandas, SciPy
- Matplotlib and Seaborn
- scikit-learn
- mlxtend for Apriori/FP-Growth where appropriate
- NLTK and/or spaCy for text preprocessing
- WordCloud for text visualization
- Git and GitHub
- Overleaf

5. 15-Week / 30-Class Master Schedule

Week	Class	Laboratory / Evaluation Topic	Focus	CLO
1	1	Python Fundamentals and Notebook Setup	Python syntax, variables, data types, control flow, functions, files, notebook discipline	CLO1
1	2	NumPy for Numerical Computing	Arrays, indexing, slicing, vectorization, statistics, reshaping	CLO1
2	1	Pandas and DataFrames	Series, DataFrame, CSV import/export, selection, filtering, grouping	CLO1,CLO2
2	2	Data Visualization and Statistical Description	Descriptive statistics, distributions, plots, correlation	CLO1
3	1	Data Quality and Cleaning	Missing values, duplicates, inconsistent values, outliers	CLO2
3	2	Data Transformation, Normalization and Encoding	Scaling, normalization, categorical encoding, train/test split	CLO2
4	1	Similarity and Dissimilarity Measures	Euclidean, Manhattan, cosine similarity, distance matrices	CLO1,CLO3
4	2	EDA and Statistical Analysis Laboratory	Integrated EDA, distributions, relationships, interpretation	CLO1,CLO2
5	1	Apriori: Frequent Itemset Mining	Support, confidence, lift, frequent itemsets	CLO3
5	2	FP-Growth: Frequent Pattern	FP-tree, mining frequent patterns, Apriori vs	CLO3,CLO4

		Mining	FP-Growth	
6	1	Simple Linear Regression	Least squares, prediction, residuals, MAE/MSE/R ²	CLO3
6	2	Multiple Linear Regression	Multiple predictors, coefficient interpretation, evaluation	CLO3
7	1	Decision Tree Classification	Entropy, information gain, tree construction, prediction	CLO3
7	2	Support Vector Machine	Margins, kernels, hyperparameters, classification	CLO3,CLO4
8	1	K-Nearest Neighbors	Distance-based classification, k selection, scaling	CLO3
8	2	Naive Bayes Classification	Bayes theorem, conditional independence, text-oriented use	CLO3
9	1	Artificial Neural Network	Perceptron/MLP, activation, forward pass, training, evaluation	CLO3
9	2	Neural Network Implementation and Comparison	MLP implementation, tuning, comparison with classical model	CLO3,CLO4
10	1	K-Means Clustering	Centroids, assignment/update, inertia, clustering workflow	CLO3
10	2	K-Medoids Clustering	Medoids, robustness, comparison with K-Means	CLO3,CLO4
11	1	DBSCAN and Density-Based Clustering	Core/border/noise points, eps, min_samples	CLO3,CLO4
11	2	Random Forest and Ensemble Learning	Bagging, decision-tree ensembles, feature importance	CLO3,CLO4
12	1	Model Evaluation and Comparison	Confusion matrix, precision, recall, F1, ROC-AUC, cross-validation	CLO3
12	2	Text Data EDA and Preprocessing	Text statistics, cleaning, tokenization, stopwords, stemming/lemmatization	CLO1,CLO2
13	1	Text Vectorization and Final Project Readiness	TF-IDF, feature inspection, baseline design, project evidence checklist	CLO2,CLO3,CLO5
13	2	FINAL PROJECT EVALUATION 1 - Problem Definition & Ethical Consideration	SEE component: problem, motivation, dataset, ethics, legal considerations	CLO4
14	1	FINAL PROJECT EVALUATION 2 - Preprocessing & EDA	SEE component: data cleaning, preprocessing, EDA and evidence	CLO1,CLO2
14	2	FINAL PROJECT EVALUATION 3 - Model Implementation	SEE component: at least two algorithms, baseline, metrics and comparison	CLO3
15	1	FINAL PROJECT EVALUATION 4 - Final Report & GitHub	SEE component: paper quality, structure, references, figures and repository	CLO5
15	2	FINAL PROJECT EVALUATION 5 - Presentation & Final Defense	SEE component: five-minute presentation and repository demonstration	CLO5

Week 1 - Class 1: Python Fundamentals and Notebook Setup

Purpose. Establish the minimum Python skills required for independent machine-learning laboratory work.

Learning Objectives

- Use variables and common Python data types.
- Use lists, tuples, dictionaries and sets.
- Write conditions, loops and functions.
- Create a structured Jupyter/Colab notebook.

Background / Theory

Python is the programming foundation for the subsequent NumPy, Pandas and scikit-learn laboratories. Students should be able to manipulate basic data structures and write small reusable functions before moving to numerical arrays.

Laboratory Procedure

1. Create a notebook with Objective, Learning Outcomes, Imports, Experiments, Results and Reflection.
2. Declare numeric, string and Boolean variables and inspect their types.
3. Create and manipulate lists, tuples, dictionaries and sets.
4. Write if/elif/else conditions and for/while loops.
5. Write functions for mean, maximum, minimum and range.
6. Read a small CSV file using Python's csv module.
7. Record the Python version and package versions.

Student Task / Extension

Create a reusable Python starter notebook and add five self-designed test cases.

Expected Output

- Clean .ipynb file
- Working functions
- CSV-reading demonstration
- Version information

Viva / Discussion Questions

- Why are functions useful in ML notebooks?
- What is the difference between a list and a NumPy array?
- Why is reproducibility important?

CLO Mapping: CLO1

Submission convention: Week1_Class1_Python_Fundamentals_and_Notebook_Setup.ipynb

Week 1 - Class 2: NumPy for Numerical Computing

Purpose. Build the numerical foundation required for statistics, regression, clustering and neural networks.

Learning Objectives

- Create one- and multi-dimensional arrays.
- Use shape, ndim, size, dtype and astype.
- Perform indexing, slicing, reshaping and vectorized operations.
- Compute common statistical summaries.

Background / Theory

NumPy arrays support efficient numerical operations and are used throughout machine-learning pipelines.

Laboratory Procedure

8. Create 1D, 2D and 3D arrays.
9. Inspect shape, dimension, size and data type.
10. Perform indexing and slicing.
11. Reshape arrays and flatten matrices.
12. Apply vectorized arithmetic.
13. Calculate mean, standard deviation, minimum, maximum, sum, argmin and argmax.
14. Compare a vectorized operation with a Python loop.

Student Task / Extension

Implement a small numerical-analysis utility using NumPy only.

Expected Output

- Array examples
- Summary statistics
- Reshaping demonstration
- Vectorization comparison

Viva / Discussion Questions

- Why is vectorization faster or more convenient?
- What is the difference between shape and size?
- Why might dtype matter?

CLO Mapping: CLO1

Submission convention: Week1_Class2_NumPy_for_Numerical_Computing.ipynb

Week 2 - Class 1: Pandas and DataFrames

Purpose. Develop practical tabular-data manipulation skills.

Learning Objectives

- Create Series and DataFrames.
- Load CSV data.
- Select, filter and sort records.
- Group and summarize data.

Background / Theory

Pandas provides the main tabular-data interface used in data mining and machine learning preprocessing.

Laboratory Procedure

15. Load a CSV dataset.
16. Inspect head, tail, shape, dtypes and describe().
17. Select columns and rows with loc/iloc.
18. Filter records using conditions.
19. Sort values.
20. Group by one or more attributes.
21. Create a summarized table and export it.

Student Task / Extension

Perform a compact exploratory report on one tabular dataset.

Expected Output

- Dataset profile
- Filtered tables
- Grouped summaries
- Exported CSV

Viva / Discussion Questions

- When should loc be used instead of iloc?
- Why inspect dtypes before modeling?
- What does groupby provide?

CLO Mapping: CLO1,CLO2

Submission convention: Week2_Class1_Pandas_and_DataFrames.ipynb

Week 2 - Class 2: Data Visualization and Statistical Description

Purpose. Connect statistical summaries to visual evidence.

Learning Objectives

- Compute mean, median, mode, variance and standard deviation.
- Generate histograms, boxplots, bar charts and scatterplots.
- Interpret distribution and correlation.

Background / Theory

Statistical descriptions and visualization reveal central tendency, spread, skewness, outliers and relationships that influence later modeling choices.

Laboratory Procedure

22. Select numerical and categorical variables.
23. Compute descriptive statistics.
24. Plot a histogram and boxplot.
25. Plot categorical frequency.
26. Create a scatterplot for two numerical variables.
27. Compute a correlation matrix and visualize it.
28. Write observations separately from interpretations.

Student Task / Extension

Produce an EDA page containing at least four complementary plots and a short interpretation.

Expected Output

- Descriptive-statistics table
- Four plots
- Correlation matrix
- Interpretation paragraph

Viva / Discussion Questions

- Does correlation imply causation?
- What does a boxplot reveal?
- When is a histogram preferable to a bar chart?

CLO Mapping: CLO1

Submission convention: Week2_Class2_Data_Visualization_and_Statistical_Description.ipynb

Week 3 - Class 1: Data Quality and Cleaning

Purpose. Prepare imperfect real-world data for analysis.

Learning Objectives

- Identify missing values, duplicates and inconsistent values.
- Detect obvious outliers.
- Apply justified cleaning operations.
- Document data-quality decisions.

Background / Theory

Data preprocessing improves reliability. Cleaning must be justified rather than performed mechanically.

Laboratory Procedure

29. Inspect missing-value counts.
30. Identify duplicate records.
31. Check inconsistent categorical labels.
32. Use suitable strategies for missing numerical and categorical values.
33. Detect outliers with IQR or a boxplot.
34. Compare the dataset before and after cleaning.
35. Record every modification.

Student Task / Extension

Create a data-quality report and cleaned version of the dataset.

Expected Output

- Missing-value table
- Duplicate report
- Outlier evidence
- Cleaned dataset

Viva / Discussion Questions

- Why should missing values not always be replaced by zero?
- What can happen if outliers are removed without justification?

CLO Mapping: CLO2

Submission convention: Week3_Class1_Data_Quality_and_Cleaning.ipynb

Week 3 - Class 2: Data Transformation, Normalization and Encoding

Purpose. Transform heterogeneous attributes into model-ready features.

Learning Objectives

- Apply min-max scaling and standardization.
- Encode categorical variables.
- Explain why scaling affects distance-based algorithms.
- Perform a correct train/test split.

Background / Theory

Many algorithms are sensitive to feature scale. Encoding converts categorical information into numerical representations.

Laboratory Procedure

36. Separate features and target.
37. Split data into training and testing subsets.
38. Fit a scaler on training data only.
39. Transform training and test data.
40. Encode categorical variables.
41. Compare feature ranges before and after scaling.
42. Demonstrate the effect of scaling on a distance calculation.

Student Task / Extension

Prepare a reusable preprocessing pipeline and explain why fitting preprocessing on the full dataset can cause leakage.

Expected Output

- Preprocessed feature matrix
- Train/test split
- Before/after scale comparison
- Leakage explanation

Viva / Discussion Questions

- What is data leakage?
- Why must a scaler be fitted on training data?
- Which algorithms are especially sensitive to feature scale?

CLO Mapping: CLO2

Submission convention: Week3_Class2_Data_Transformation_Normalization_and-Encoding.ipynb

Week 4 - Class 1: Similarity and Dissimilarity Measures

Purpose. Introduce distance and similarity measures used in classification and clustering.

Learning Objectives

- Calculate Euclidean and Manhattan distance.
- Calculate cosine similarity.
- Construct a small distance matrix.
- Relate distance choice to data type.

Background / Theory

Similarity and dissimilarity provide the mathematical basis for KNN, clustering and text-vector comparison.

Laboratory Procedure

43. Create a small numerical dataset.
44. Calculate Euclidean distance manually for selected pairs.
45. Verify with SciPy/sklearn utilities.
46. Calculate Manhattan distance.
47. Calculate cosine similarity.
48. Build and inspect a pairwise distance matrix.
49. Discuss how feature scaling changes distances.

Student Task / Extension

Compare at least three measures on the same sample pairs and explain differences.

Expected Output

- Distance calculations
- Similarity matrix
- Comparison table

Viva / Discussion Questions

- Why is cosine similarity useful for text vectors?
- How does scaling change Euclidean distance?

CLO Mapping: CLO1,CLO3

Submission convention: Week4_Class1_Similarity_and_Dissimilarity_Measures.ipynb

Week 4 - Class 2: EDA and Statistical Analysis Laboratory

Purpose. Integrate the statistical and visualization techniques into a complete data-analysis workflow.

Learning Objectives

- Conduct structured EDA.
- Identify distributions, relationships, outliers and imbalance.
- Translate observations into modeling considerations.

Background / Theory

EDA is not merely plotting; it is the process of understanding the dataset sufficiently to justify preprocessing and modeling choices.

Laboratory Procedure

50. Load a new dataset.
51. Profile columns and data types.
52. Compute descriptive statistics.
53. Analyze missingness and class distribution.
54. Generate appropriate univariate and bivariate plots.
55. Analyze numerical correlations.
56. Write a concise findings section linking EDA to later preprocessing.

Student Task / Extension

Produce an instructor-checkable EDA notebook with evidence-based conclusions.

Expected Output

- Complete EDA notebook
- Dataset summary
- Visualizations
- Findings and modeling implications

Viva / Discussion Questions

- Which EDA finding would cause you to change preprocessing?
- How can class imbalance appear visually?

CLO Mapping: CLO1,CLO2

Submission convention: Week4_Class2_EDA_and_Statistical_Analysis_Laboratory.ipynb

Week 5 - Class 1: Apriori: Frequent Itemset Mining

Purpose. Implement association-rule mining using the Apriori principle.

Learning Objectives

- Explain support, confidence and lift.
- Generate frequent itemsets.
- Generate association rules.
- Interpret useful and misleading rules.

Background / Theory

Apriori repeatedly generates and tests candidate itemsets. Support measures occurrence, confidence measures conditional occurrence and lift compares a rule with independence.

Laboratory Procedure

57. Load a transaction dataset.
58. Convert transactions into a one-hot representation.
59. Choose minimum support.
60. Generate frequent itemsets using Apriori.
61. Generate rules using confidence.
62. Filter rules using lift.
63. Sort and interpret the strongest rules.

Student Task / Extension

Implement a small support-count calculation from scratch before verifying with mlxtend.

Expected Output

- Frequent-itemset table
- Association-rule table
- Interpretation of top rules

Viva / Discussion Questions

- Why can high confidence be misleading?
- What does lift greater than 1 suggest?

CLO Mapping: CLO3

Submission convention: Week5_Class1_Apriori_Frequent_Itemset_Mining.ipynb

Week 5 - Class 2: FP-Growth: Frequent Pattern Mining

Purpose. Study an efficient alternative to Apriori and compare the two approaches.

Learning Objectives

- Explain the FP-tree idea.
- Mine frequent patterns.
- Compare computational logic with Apriori.
- Interpret association rules.

Background / Theory

FP-Growth compresses transactions into an FP-tree and avoids repeated candidate generation.

Laboratory Procedure

64. Prepare the same transaction dataset used for Apriori.
65. Generate frequent patterns with FP-Growth.
66. Generate association rules.
67. Compare rule quality with Apriori.
68. Record runtime or candidate-generation differences where practical.
69. Discuss when FP-Growth is preferable.

Student Task / Extension

Create a concise Apriori-versus-FP-Growth comparison.

Expected Output

- FP-Growth itemsets
- Rules
- Comparison table
- Interpretation

Viva / Discussion Questions

- What problem of Apriori does FP-Growth address?
- What information is stored in an FP-tree?

CLO Mapping: CLO3,CLO4

Submission convention: Week5_Class2_FP_Growth_Frequent_Pattern_Mining.ipynb

Week 6 - Class 1: Simple Linear Regression

Purpose. Implement and evaluate a basic predictive regression model.

Learning Objectives

- Explain the linear regression equation.
- Fit a simple regression model.
- Interpret coefficients.
- Evaluate using MAE, MSE and R^2 .

Background / Theory

Simple linear regression estimates a continuous target from one predictor by minimizing squared residual error.

Laboratory Procedure

70. Select one predictor and continuous target.
71. Plot predictor versus target.
72. Fit a linear regression model.
73. Calculate predictions and residuals.
74. Compute MAE, MSE and R^2 .
75. Plot regression line and residual evidence.
76. Compare the library result with a small from-scratch calculation.

Student Task / Extension

Explain whether the relationship appears suitable for linear prediction.

Expected Output

- Regression equation
- Prediction plot
- Residual analysis
- Metrics

Viva / Discussion Questions

- What does R^2 measure?
- Why is a residual important?
- Can high R^2 prove causation?

CLO Mapping: CLO3

Submission convention: Week6_Class1_Simple_Linear_Regression.ipynb

Week 6 - Class 2: Multiple Linear Regression

Purpose. Extend regression to multiple predictors and evaluate model behavior.

Learning Objectives

- Fit multiple linear regression.
- Interpret coefficients carefully.
- Compare one- and multi-feature models.
- Identify possible multicollinearity.

Background / Theory

Multiple regression models a continuous target using several predictors. Correlated predictors can complicate coefficient interpretation.

Laboratory Procedure

77. Prepare numerical features.
78. Split the data.
79. Fit simple and multiple regression models.
80. Compare MAE, MSE and R^2 .
81. Inspect coefficients.
82. Calculate feature correlations.
83. Discuss possible multicollinearity and limitations.

Student Task / Extension

Prepare a comparison table showing whether additional features improve predictive performance.

Expected Output

- Two regression models
- Metric comparison
- Coefficient table
- Discussion

Viva / Discussion Questions

- Why can adding features hurt generalization?
- What is multicollinearity?

CLO Mapping: CLO3

Submission convention: Week6_Class2_Multiple_Linear_Regression.ipynb

Week 7 - Class 1: Decision Tree Classification

Purpose. Understand and implement a tree-based classifier.

Learning Objectives

- Explain entropy and information gain.
- Build a Decision Tree classifier.
- Visualize a tree.
- Evaluate predictions.

Background / Theory

Decision Trees recursively split data using a criterion such as entropy/information gain or Gini impurity.

Laboratory Procedure

84. Prepare a classification dataset.
85. Calculate a small entropy example manually.
86. Train a Decision Tree.
87. Visualize the tree.
88. Generate predictions.
89. Create a confusion matrix.
90. Calculate precision, recall and F1-score.

Student Task / Extension

Compare a shallow tree with a deeper tree and discuss overfitting.

Expected Output

- Tree visualization
- Confusion matrix
- Classification metrics
- Depth comparison

Viva / Discussion Questions

- Why can a very deep tree overfit?
- What is information gain?

CLO Mapping: CLO3

Submission convention: Week7_Class1_Decision_Tree_Classification.ipynb

Week 7 - Class 2: Support Vector Machine

Purpose. Implement SVM classification and examine margins and kernels.

Learning Objectives

- Explain the separating hyperplane and margin.
- Train a linear and kernel SVM.
- Tune C and gamma conceptually.
- Evaluate performance.

Background / Theory

SVM seeks a decision boundary that maximizes the margin. Kernels allow nonlinear decision boundaries.

Laboratory Procedure

91. Scale the features.
92. Train a linear SVM.
93. Train an RBF-kernel SVM.
94. Compare parameter settings.
95. Evaluate using precision, recall and F1.
96. Plot a 2D decision boundary where feasible.
97. Record the effect of scaling.

Student Task / Extension

Produce a parameter-performance table and justify the selected SVM configuration.

Expected Output

- Model comparison
- Metrics
- Decision-boundary plot where possible

Viva / Discussion Questions

- Why is scaling important for SVM?
- What does C control conceptually?

CLO Mapping: CLO3,CLO4

Submission convention: Week7_Class2_Support_Vector_Machine.ipynb

Week 8 - Class 1: K-Nearest Neighbors

Purpose. Apply distance-based classification and investigate the role of k .

Learning Objectives

- Explain nearest-neighbor classification.
- Apply KNN with multiple k values.
- Understand the role of feature scaling.
- Select k using validation evidence.

Background / Theory

KNN predicts based on nearby training examples, making distance and feature scaling central to its performance.

Laboratory Procedure

98. Prepare and scale a dataset.
99. Train KNN for several k values.
100. Evaluate each setting.
101. Plot validation performance against k .
102. Inspect a confusion matrix.
103. Explain underfitting versus overfitting as k changes.

Student Task / Extension

Select a final k based on evidence rather than guesswork.

Expected Output

- k -performance plot
- Best- k rationale
- Confusion matrix
- Metrics

Viva / Discussion Questions

- What happens when $k=1$?
- Why does KNN require scaling?

CLO Mapping: CLO3

Submission convention: Week8_Class1_K_Nearest_Neighbors.ipynb

Week 8 - Class 2: Naive Bayes Classification

Purpose. Implement probabilistic classification and connect it to text-oriented problems.

Learning Objectives

- Explain Bayes theorem.
- Understand the conditional-independence assumption.
- Train a Naive Bayes classifier.
- Evaluate classification performance.

Background / Theory

Naive Bayes estimates class probabilities using Bayes theorem and assumes conditional independence between features given the class.

Laboratory Procedure

104. Prepare a classification dataset.
105. Train an appropriate Naive Bayes variant.
106. Generate class probabilities.
107. Evaluate with precision, recall and F1.
108. Inspect a confusion matrix.
109. Discuss the conditional-independence assumption.
110. Explain why Naive Bayes is common in text classification.

Student Task / Extension

Compare Naive Bayes with one previous classifier on the same dataset.

Expected Output

- Probability outputs
- Metrics
- Confusion matrix
- Comparison paragraph

Viva / Discussion Questions

- What is the naive assumption?
- Why can Naive Bayes work well for sparse text features?

CLO Mapping: CLO3

Submission convention: Week8_Class2_Naive_Bayes_Classification.ipynb

Week 9 - Class 1: Artificial Neural Network

Purpose. Introduce a practical multilayer neural-network workflow.

Learning Objectives

- Explain neurons, weights and activation functions.
- Describe forward propagation and loss.
- Train an MLP classifier.
- Evaluate performance.

Background / Theory

A neural network combines weighted transformations and nonlinear activation functions. Training adjusts weights to reduce loss.

Laboratory Procedure

111. Prepare and scale a classification dataset.
112. Define an MLP architecture.
113. Train the network.
114. Record training and validation performance.
115. Evaluate on the test set.
116. Plot a learning curve.
117. Discuss overfitting and regularization.

Student Task / Extension

Modify one architecture parameter and compare performance.

Expected Output

- Network configuration
- Learning curve
- Test metrics
- Architecture comparison

Viva / Discussion Questions

- Why are nonlinear activations needed?
- What is overfitting in a neural network?

CLO Mapping: CLO3

Submission convention: Week9_Class1_Artificial_Neural_Network.ipynb

Week 9 - Class 2: Neural Network Implementation and Comparison

Purpose. Evaluate neural networks against classical models rather than treating them as automatically superior.

Learning Objectives

- Tune a small MLP.
- Compare MLP with a classical classifier.
- Interpret training behavior.
- Select a model based on evidence.

Background / Theory

Model selection depends on data characteristics and empirical evaluation. A more complex model is not automatically better.

Laboratory Procedure

118. Train an MLP with two configurations.
119. Train one classical baseline.
120. Compare accuracy, precision, recall and F1.
121. Plot training history.
122. Inspect errors.
123. Record computational considerations.
124. Write a model-selection conclusion.

Student Task / Extension

Create a model comparison table and identify the most defensible model for the dataset.

Expected Output

- Comparison table
- Learning curves
- Error examples
- Model-selection rationale

Viva / Discussion Questions

- Why might a simpler model be preferable?
- How do you distinguish training and validation performance?

CLO Mapping: CLO3,CLO4

Submission convention: Week9_Class2_Neural_Network_Implementation_and_Comparison.ipynb

Week 10 - Class 1: K-Means Clustering

Purpose. Implement centroid-based clustering and evaluate cluster quality.

Learning Objectives

- Explain the K-Means objective.
- Run K-Means.
- Use inertia/elbow analysis.
- Interpret cluster assignments.

Background / Theory

K-Means iteratively assigns points to the nearest centroid and updates centroids to minimize within-cluster squared distance.

Laboratory Procedure

125. Scale suitable features.
126. Run K-Means for multiple k values.
127. Record inertia.
128. Plot an elbow curve.
129. Visualize clusters in two dimensions where possible.
130. Interpret cluster characteristics.
131. Discuss sensitivity to initialization.

Student Task / Extension

Select k using an evidence-based argument.

Expected Output

- Cluster assignments
- Elbow plot
- Cluster visualization
- Interpretation

Viva / Discussion Questions

- Why does K-Means need k?
- What does inertia measure?

CLO Mapping: CLO3

Submission convention: Week10_Class1_K_Means_Clustering.ipynb

Week 10 - Class 2: K-Medoids Clustering

Purpose. Compare medoid-based clustering with K-Means.

Learning Objectives

- Explain medoids.
- Run K-Medoids.
- Compare robustness and representatives.
- Evaluate clusters.

Background / Theory

K-Medoids selects actual observations as cluster representatives and can be more robust to outliers than centroid-based methods.

Laboratory Procedure

132. Prepare and scale data.
133. Run K-Medoids for selected k values.
134. Inspect medoid observations.
135. Compare with K-Means.
136. Evaluate cluster separation where applicable.
137. Discuss outlier influence.

Student Task / Extension

Produce a K-Means versus K-Medoids comparison table.

Expected Output

- Medoids
- Cluster comparison
- Evaluation evidence
- Interpretation

Viva / Discussion Questions

- What makes a medoid different from a centroid?
- Why can medoids be more interpretable?

CLO Mapping: CLO3,CLO4

Submission convention: Week10_Class2_K_Medoids_Clustering.ipynb

Week 11 - Class 1: DBSCAN and Density-Based Clustering

Purpose. Use density-based clustering to identify arbitrary-shaped clusters and noise.

Learning Objectives

- Explain eps and min_samples.
- Identify core, border and noise points.
- Run DBSCAN.
- Evaluate and interpret clusters.

Background / Theory

DBSCAN groups points according to density and can label isolated observations as noise without requiring the number of clusters in advance.

Laboratory Procedure

138. Scale the data.
139. Experiment with eps and min_samples.
140. Run DBSCAN.
141. Visualize cluster labels.
142. Identify noise points.
143. Compare with K-Means.
144. Discuss sensitivity to parameter choices.

Student Task / Extension

Create a parameter comparison showing how cluster structure changes.

Expected Output

- Cluster plot
- Noise identification
- Parameter comparison
- Interpretation

Viva / Discussion Questions

- What is a core point?
- Why can DBSCAN discover noise?
- Why is parameter selection difficult?

CLO Mapping: CLO3,CLO4

Submission convention: Week11_Class1_DBSCAN_and_Density_Based_Clustering.ipynb

Week 11 - Class 2: Random Forest and Ensemble Learning

Purpose. Understand ensemble learning and compare Random Forest with a single Decision Tree.

Learning Objectives

- Explain bagging and random feature selection.
- Train a Random Forest.
- Inspect feature importance.
- Compare with a Decision Tree.

Background / Theory

Random Forest combines many decision trees trained using bootstrap samples and randomized feature subsets, generally improving stability and generalization.

Laboratory Procedure

145. Prepare a classification dataset.
146. Train a Decision Tree baseline.
147. Train a Random Forest.
148. Compare metrics.
149. Inspect feature importance.
150. Change the number of trees.
151. Discuss variance reduction and interpretability.

Student Task / Extension

Write a defensible explanation of why the ensemble does or does not improve over the single tree.

Expected Output

- Metric comparison
- Feature-importance plot
- Parameter comparison

Viva / Discussion Questions

- Why does bagging help?
- Why can Random Forest be less interpretable than one tree?

CLO Mapping: CLO3,CLO4

Submission convention: Week11_Class2_Random_Forest_and_Ensemble_Learning.ipynb

Week 12 - Class 1: Model Evaluation and Comparison

Purpose. Build the evaluation skills needed for responsible model selection and the final project.

Learning Objectives

- Construct confusion matrices.
- Calculate accuracy, precision, recall and F1.
- Plot ROC and calculate AUC for binary classification.
- Use cross-validation appropriately.

Background / Theory

Evaluation must match the task and data distribution. Accuracy alone can be misleading under class imbalance.

Laboratory Procedure

152. Train two or more classifiers.
153. Generate confusion matrices.
154. Calculate per-class precision, recall and F1.
155. Plot ROC curves where appropriate.
156. Calculate AUC.
157. Perform stratified cross-validation.
158. Prepare a model comparison table.
159. Write a short evidence-based model-selection statement.

Student Task / Extension

Repeat the analysis under an imbalanced-data scenario and explain which metric becomes more informative.

Expected Output

- Evaluation table
- Confusion matrices
- ROC-AUC
- Cross-validation results
- Model-selection statement

Viva / Discussion Questions

- Why can accuracy be misleading?
- What does recall emphasize?
- When is ROC-AUC useful?

CLO Mapping: CLO3

Submission convention: Week12_Class1_Model_Evaluation_and_Comparison.ipynb

Week 12 - Class 2: Text Data EDA and Preprocessing

Purpose. Introduce the text-analysis workflow required for the final project.

Learning Objectives

- Load text data into Pandas.
- Analyze text length and class balance.
- Clean text.
- Tokenize and remove stopwords.
- Apply stemming or lemmatization.

Background / Theory

The project manual requires text-only data and identifies cleaning, tokenization, stopword removal and stemming/lemmatization as core preprocessing operations.

Laboratory Procedure

160. Load a text dataset.
161. Inspect columns and missing values.
162. Calculate text length distributions.
163. Analyze class balance where applicable.
164. Create a word-frequency or word-cloud visualization.
165. Convert text to lowercase.
166. Remove punctuation and unwanted characters.
167. Tokenize.
168. Remove stopwords.
169. Apply stemming or lemmatization.
170. Compare raw and cleaned examples.

Student Task / Extension

Produce a cleaned text dataset and document every transformation.

Expected Output

- Text EDA plots
- Class-balance table
- Before/after cleaning examples
- Cleaned text column

Viva / Discussion Questions

- Why should preprocessing be applied consistently?
- Can removing stopwords ever remove useful information?

CLO Mapping: CLO1,CLO2

Submission convention: Week12_Class2_Text_Data_EDA_and_Preprocessing.ipynb

Week 13 - Class 1: Text Vectorization and Final Project Readiness

Purpose. Convert cleaned text into numerical features and verify that students are ready for the final project assessment.

Learning Objectives

- Build TF-IDF features.
- Inspect vocabulary and matrix dimensions.
- Create a simple baseline.
- Prepare an experiment record containing the required project evidence.

Background / Theory

TF-IDF represents terms according to their frequency within a document and their inverse frequency across documents. The project requires at least two algorithms and a baseline for classification.

Laboratory Procedure

171. Vectorize cleaned text using TF-IDF.
172. Inspect vocabulary size and sample features.
173. Measure matrix dimensions and sparsity.
174. Create a simple baseline.
175. Train a quick validation model.
176. Record the evaluation metrics.
177. Create a project evidence checklist: problem, ethics, preprocessing/EDA, models, paper, GitHub and presentation.
178. Verify that project notebooks and figures can be reproduced.

Student Task / Extension

Complete a final project readiness checklist. This is preparation, not final evaluation.

Expected Output

- TF-IDF matrix
- Vocabulary inspection
- Baseline result
- Readiness checklist

Viva / Discussion Questions

- Why is TF-IDF suitable for sparse text?
- What is a baseline?
- What evidence should be retained for the final evaluation?

CLO Mapping: CLO2,CLO3,CLO5

Submission convention: Week13_Class1_Text_Vectorization_and_Final_Project_Readiness.ipynb

Week 13 - Class 2: FINAL PROJECT EVALUATION 1 - Problem Definition & Ethical Consideration

Purpose. Evaluate the project's problem formulation and responsible-use considerations. This class is assessment-only.

Learning Objectives

- Clearly state the problem and motivation.
- Identify the dataset source and intended task.
- Explain potential harm, bias, privacy and ethical/legal concerns.
- Connect the problem to a meaningful societal context.

Background / Theory

The project-format document requires a clear problem statement, dataset source and potential societal impact, together with ethical checks concerning harm, bias, consent and personally identifiable information.

Laboratory Procedure

179. Student presents the problem statement and motivation.
180. Student identifies the dataset source and task.
181. Student explains whether the dataset contains potential bias or non-consensual/PII content.
182. Student discusses who may benefit and who may be harmed.
183. Student explains relevant ethical/legal concerns.
184. Instructor evaluates the evidence against the 5-mark component.

Student Task / Extension

Respond to evaluator questions without reading directly from the document or slides.

Expected Output

- Problem statement
- Dataset identification
- Ethical-risk analysis
- Societal-impact consideration

Viva / Discussion Questions

- What is the exact problem your model solves?
- Who could be harmed by a wrong prediction?
- What privacy or bias issue is most important in your dataset?

CLO Mapping: 5 marks - Problem definition & ethical consideration - CLO4

Submission convention:

Week13_Class2_FINAL_PROJECT_EVALUATION_1_Problem_Definition_Ethical_Consideration.ipynb

Week 14 - Class 1: FINAL PROJECT EVALUATION 2 - Data Preprocessing & Exploratory Data Analysis

Purpose. Evaluate whether the student correctly transformed the raw text dataset into a reliable model-ready dataset.

Learning Objectives

- Demonstrate data cleaning.
- Demonstrate text preprocessing.
- Present EDA findings.
- Explain class balance and relevant statistics.
- Justify preprocessing decisions.

Background / Theory

The project manual requires loading text data into Pandas, analyzing text lengths, class balance and frequent words, and performing mandatory text-cleaning operations.

Laboratory Procedure

185. Student opens the project notebook.
186. Demonstrate dataset source and raw-data structure.
187. Show missing/duplicate/inconsistent-data handling where relevant.
188. Show text-length and class-distribution analysis.
189. Show frequent-word or word-cloud evidence.
190. Demonstrate lowercase conversion, punctuation/special-character handling, tokenization, stopword removal and stemming/lemmatization as applicable.
191. Explain how train/test separation prevents leakage.
192. Instructor evaluates the evidence against the 10-mark component.

Student Task / Extension

Answer questions about why each preprocessing operation was selected and what information it may remove.

Expected Output

- Raw-vs-cleaned evidence
- EDA figures
- Preprocessing pipeline
- Class/distribution analysis

Viva / Discussion Questions

- Why did you choose this preprocessing method?
- How did you prevent data leakage?
- What does your EDA tell you about the modeling problem?

CLO Mapping: 10 marks - Data preprocessing & exploratory data analysis - CLO1, CLO2

Submission convention:

Week14_Class1_FINAL_PROJECT_EVALUATION_2_Data_Preprocessing_Exploratory_Data_Analysis.ipynb

Week 14 - Class 2: FINAL PROJECT EVALUATION 3 - Model Implementation

Purpose. Evaluate the technical implementation and comparison of the machine-learning models.

Learning Objectives

- Implement at least two algorithms.
- Include a simple baseline.
- Evaluate using appropriate metrics.
- Compare models and justify the selected model.
- Demonstrate reproducible execution.

Background / Theory

The project manual requires at least two algorithms and a baseline. Suitable metrics include accuracy, precision, recall and F1-score, with confusion matrix and ROC curve for binary classification where appropriate.

Laboratory Procedure

193. Student runs or demonstrates the final notebook.
194. Identify the baseline.
195. Demonstrate at least two implemented algorithms.
196. Explain model settings and major hyperparameters.
197. Show evaluation metrics.
198. Show confusion matrix.
199. Show ROC curve/AUC where appropriate.
200. Compare models in a single table.
201. Defend the final model choice.
202. Instructor evaluates the evidence against the 15-mark component.

Student Task / Extension

Explain one major error case and one limitation of the selected model.

Expected Output

- At least two trained models
- Baseline
- Evaluation table
- Confusion matrix
- ROC/AUC where applicable
- Model-selection rationale

Viva / Discussion Questions

- Why did your best model outperform the baseline?
- Which metric mattered most and why?
- What is your model's most important limitation?

CLO Mapping: 15 marks - Model implementation (at least two algorithms) - CLO3

Submission convention:

Week14_Class2_FINAL_PROJECT_EVALUATION_3_Model_Implementation.ipynb

Week 15 - Class 1: FINAL PROJECT EVALUATION 4 - IEEE Paper & GitHub Repository

Purpose. Evaluate the quality and completeness of the final written research artifact and repository.

Learning Objectives

- Present a coherent IEEE-style paper.
- Use appropriate references.
- Include required tables and figures.
- Document methodology and results clearly.
- Maintain a usable GitHub repository.

Background / Theory

The project format specifies a 4–6 page IEEE-style paper with Abstract, Introduction, Related Work, Methodology, Experiments & Results, Discussion, Conclusion and References. It also requires dataset/model tables and key figures.

Laboratory Procedure

203. Instructor checks paper structure.
204. Check abstract, introduction and stated contributions.
205. Check related work and scholarly references.
206. Check methodology, preprocessing and model descriptions.
207. Check dataset-statistics and model-comparison tables.
208. Check required figures such as word cloud/top features, confusion matrix and ROC where appropriate.
209. Check discussion of bias, fairness, limitations and future work.
210. Check GitHub README, setup instructions, notebook and repository organization.
211. Evaluate against the 10-mark component.

Student Task / Extension

Student answers questions about where the main contribution, evidence and limitations are located in the paper.

Expected Output

- IEEE paper PDF
- Tables and figures
- Reference list
- GitHub repository with README and runnable notebook

Viva / Discussion Questions

- What is your main contribution?
- Which figure provides your strongest evidence?
- How can another person reproduce your work from GitHub?

CLO Mapping: 10 marks - IEEE paper (structure, clarity, references, figures) - CLO5

Submission convention:

Week15_Class1_FINAL_PROJECT_EVALUATION_4_IEEE_Paper_GitHub_Repository.ipynb

Week 15 - Class 2: FINAL PROJECT EVALUATION 5 - Presentation & Final Defense

Purpose. Conduct the final five-minute presentation and assess the student's ability to communicate and defend the project.

Learning Objectives

- Present problem, method, results and impact within five minutes.
- Explain major design choices.
- Defend results and limitations.
- Demonstrate repository consistency.

Background / Theory

The project format requires a five-minute live presentation together with the GitHub repository. The presentation should be consistent with the final notebook and paper.

Laboratory Procedure

212. Student presents problem and motivation.
213. State dataset and preprocessing briefly.
214. Present the two or more models and key results.
215. Show the most important evaluation evidence.
216. Explain societal/ethical impact and limitations.
217. Demonstrate or identify the GitHub repository.
218. Answer evaluator questions.
219. Instructor evaluates communication, technical consistency and repository evidence against the 10-mark component.

Student Task / Extension

Presentation should be concise, evidence-based and understandable without reading the slides verbatim.

Expected Output

- Five-minute presentation
- Repository link
- Consistent notebook/paper/results
- Viva responses

Viva / Discussion Questions

- Why should your result be trusted?
- What would you change with more data or time?
- What is the biggest risk of deploying your model?

CLO Mapping: 10 marks - Presentation (5 minutes) + GitHub repository - CLO5

Submission convention:

Week15_Class2_FINAL_PROJECT_EVALUATION_5_Presentation_Final_Defense.ipynb

6. Final Project Evaluation - SEE 50 Marks

The final five laboratory classes are reserved exclusively for the final project evaluation. The five classes map directly to the five components of the SEE rubric.

Component	Marks	CLO Mapped	Evaluation Class
Problem definition & ethical consideration	5	CLO4	Class 26 - Week 13, Class 2
Data preprocessing & exploratory data analysis	10	CLO1, CLO2	Class 27 - Week 14, Class 1
Model implementation (at least two algorithms)	15	CLO3	Class 28 - Week 14, Class 2
IEEE paper (structure, clarity, references, figures)	10	CLO5	Class 29 - Week 15, Class 1
Presentation (5 minutes) + GitHub repository	10	CLO5	Class 30 - Week 15, Class 2
Total	50		

7. Final Project Minimum Requirements

- The project remains text-data focused.
- For classification, use at least 1,000 labeled text samples unless an alternative research design is explicitly approved.
- Use at least two machine-learning algorithms and a simple baseline.
- Report suitable evaluation metrics including accuracy, precision, recall and F1-score where applicable.
- Include a confusion matrix and ROC curve/AUC for binary classification where appropriate.
- Include bias/fairness analysis where meaningful subgroup information exists.
- Discuss societal and environmental impact.
- Submit a 4–6 page IEEE-style report.
- Maintain a private GitHub repository with instructor access and a README with setup instructions.
- Deliver a five-minute presentation.

8. Report Structure Required for Evaluation

Section	Required Content
Abstract	150–200 words: problem, method, key result and impact.
Introduction	Problem motivation and 3–4 concise contributions.
Related Work	10–15 scholarly references from appropriate academic sources.
Methodology	Dataset, preprocessing, vectorization, model design and important hyperparameters.
Experiments & Results	Dataset statistics, evaluation metrics, tables and figures.
Discussion	Bias, fairness, limitations, errors, ethical/legal issues and future work.
Conclusion	Concise findings and real-world impact.
References	IEEE-style references.

9. Mandatory / Expected Figures and Tables

- Dataset statistics table.
- Model comparison table containing appropriate metrics.
- Word cloud or top-feature visualization.
- Confusion matrix.

- ROC curve for binary classification where appropriate.
- All figures and tables must be numbered, captioned and discussed in the paper.

10. Final Project Evaluation Procedure

220. Students must arrive with the final notebook, paper, presentation and GitHub repository ready.
221. The evaluator checks the relevant evidence for only the component assigned to that class.
222. Students must be able to explain their own implementation and results.
223. Marks are awarded according to the stated component and CLO mapping.
224. The five evaluation classes together constitute the 50-mark SEE/project assessment.
225. No new project-development topic is introduced during Classes 26–30.

11. Final Submission Checklist

- Notebook runs end-to-end from a clean kernel without errors.
- Important code cells contain comments.
- Dataset source/link is documented.
- Cleaning and preprocessing decisions are documented.
- Train/test separation is correct and leakage is avoided.
- At least two ML models are implemented and compared against a baseline.
- Evaluation table includes suitable metrics.
- Confusion matrix is included.
- ROC curve/AUC is included when appropriate.
- Fairness/bias analysis is included where meaningful.
- Societal and environmental impact discussion is present.
- GitHub README contains setup and execution instructions.
- Report contains all required sections and references.
- Required figures and tables are numbered, captioned and discussed.
- Five-minute presentation is complete and consistent with the paper.

12. Recommended Notebook Template

226. Title and student information
227. Objective
228. Learning outcomes
229. Background / theory
230. Required libraries
231. Dataset description
232. Implementation
233. Output / visualization
234. Interpretation
235. Error analysis or limitation
236. Conclusion

237. Viva preparation

13. Instructor Implementation Note

The first 25 classes are designed to teach and prepare. The final five classes are designed to assess. This separation is intentional: students should complete their project development before entering the SEE evaluation phase. By the end of Class 25, students should already have their project problem, dataset, EDA, preprocessing, vectorization, baseline and model experiments substantially completed. The remaining five classes evaluate those artifacts rather than teaching additional project-development content.